Mohammad Azhar Khan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.24
Sri Chaitanya Techno School, Visakhapatnam	Class XII (CBSE)	2020	82.80%
ST Xavier's High School, Kantabanji	Class X (CBSE)	2018	82.60%

RESEARCH AND WORK EXPERIENCE

ScaleX Lab, IIT Hyderabad | Research Assistant

- Researching **Scalable Data Science, Theoretical ML, and Randomized Algorithms** for massive, high-dimensional datasets to improve computational and memory efficiency.
- Developing **tensor sketching and dimensionality-reduction methods** for compact, accurate representations that enable efficient processing.
- Benchmarking **accuracy, runtime, memory use, and scalability** against theoretical guarantees to guide algorithm design.

SPIRE Lab, IISc Bangalore | Research Intern

- Built and evaluated **RNN and Transformer models** for automated grammatical error detection, improving **F1-score by 8% over baseline**.
- Designed **pronunciation assessment** for the **English Gyani app**, enabling evaluation and scoring to support spoken-English practice.

PROJECTS

Melco Hospital Care (MELCO Hackathon Winner) -- *GitHub*

- **Engineered an Agentic AI healthcare framework** for public facilities that won the **"Best Idea Award"** out of nationwide teams, securing an internship and invite to Japan.
- **Architected autonomous AI agents** using LangChain and Python to streamline hospital workflow management, patient routing, and real-time medical facility diagnostics.
- **Integrated large language models with custom operational tools**, enabling agents to autonomously process facility data and significantly **reduce manual administrative overhead** for staff.

Vision-Language Learning for Product Price Prediction -- *GitHub*

- Developed a **multimodal price prediction model using OpenCLIP ViT-L/14**, combining product images and catalog descriptions from a dataset of **75,000 products**.
- Designed a **custom regression head** and selectively fine-tuned both encoders using **log-price targets, Smooth L1 loss, and mixed-precision training**.
- Achieved a **validation RMSE of 23.74** and implemented batched inference to generate **75,000 positive product price predictions**.

GPU Kernel Memory Footprint Analysis with LLVM -- *Github*

- Developed a **C++ LLVM analysis pass** connecting host and device IR to estimate memory access ranges for **CUDA kernels** without executing GPU code.
- Implemented **MemorySSA-based launch configuration recovery**, tracing memory dependencies and memcpy chains to extract grid and block dimensions from CUDA dim3 constructors.
- Combined **ScalarEvolution, NVVM intrinsic modeling, and interval arithmetic** to analyze thread-indexed load/store operations and report **per-argument read/write index ranges**.

PUBLICATIONS

Improving TensorSketch Using Complex Random Variables -- UAI 2026 (Accepted)

- Proposed **Complex-to-Real TensorSketch** for polynomial kernels, reducing estimator variance while retaining input-sparsity efficiency.
- Reduced TensorSketch's **degree-dependent estimator variance from 3^p to 2^p scaling**, delivering more accurate and stable kernel approximations on real-world benchmarks.

Efficient and Accurate Tensor Compression via Recursive Sketching -- AISTATS 2026 (Accepted)

- Proposed a **Recursive TensorSketch algorithm** for scalable inner-product estimation in high-order tensors.
- Achieved **polynomial variance dependence on tensor order** with **improved asymptotic runtime** over prior baselines.
- Delivered **substantial asymptotic runtime improvements** by exploiting structured tensor decompositions and recursive computation.

Recursive Sketching for Efficient and Accurate Tensor Compression -- JAIR (Submitted)

- Extended journal version of the AISTATS 2026 work on randomized compression of high-order tensors.

Variance Reduction in Sketching Algorithms via Complex Random Variables -- TMLR (Submitted)

- Generalized **complex randomization across CountSketch, TensorSketch, and Recursive TensorSketch** to achieve more reliable sketching.

SKILLS

Languages: Python, C++, C, Java, JavaScript

ML & AI: PyTorch, TensorFlow, Scikit-Learn, Transformers, Multimodal Learning, LLMs, Agentic AI, LangChain, LangGraph

Algorithms & Tools: Randomized Algorithms, TensorSketch, Dimensionality Reduction, LLVM, CUDA, NumPy, Pandas, SciPy

ACHIEVEMENTS

- Shortlisted among the **Top 30** teams from **200+ submissions** to represent IIT Hyderabad at **Smart India Hackathon (SIH) 2026.**
- Won the **"Best Idea Award"** at the **MELCO Hackathon 2025** by Mitsubishi Electric for building an agentic AI system for public healthcare facilities; secured an internship and an invite to Japan.
- Secured **AIR 30 in Amazon ML Challenge 2025** among 6,000+ teams by developing a custom multimodal PriceHead and gradual unfreezing strategy for large-scale price prediction.
- Achieved **Rank 54 globally** among 1,000+ entries in the **Shell AI** Fuel Blend Properties Prediction **Hackathon (2025)**.
- **Placed 3rd overall** among **150+ teams** across India in **Hack4Bengal 2.0** organized in collaboration with the Dept. of IT & Electronics, Govt. of West Bengal.
- Grand Finalist in the Smart Odisha Hackathon 2022 organized by OUTR, BBSR and the Govt. of Odisha.
- Selected for **Amazon ML Summer School '22,** finishing among the top 1,000 students globally.
- **700+** DSA Questions Solved in **GeeksForGeeks** and **LeetCode**.

POSITIONS OF RESPONSIBILITY AND VOLUNTEERING

Placement Coordinator, OCS, IITH

- Facilitate placement drives for 1,000+ students, manage corporate relations, and coordinate hybrid interview logistics.

Student Volunteer, ACM India

- Supported event operations and logistics for the national ACM ARCS & ACM India Annual Event 2026 at IIT Hyderabad.

Teaching Assistant, IIT Hyderabad

- Mentored students in **CS6713 - Scalable Algorithms for Data Analysis**, conducted office hours, and evaluated course assignments under Dr. Rameshwar Pratap.